Data science pracs 2,3,4,6,7,8

Prac 2 -

Codes-

1)Sample t test: Apple

```
apple <- read.csv(file.choose(), sep = ",", header = TRUE, stringsAsFactors = FALSE)
colnames(apple) <- trimws(colnames(apple))
summary(apple)
apple$C1 <- as.numeric(apple$C1)
t.test(apple$C1, alternative = "less", mu = 97)
```

1).CSV file to use-

1 sample t test _apple - 1 sample t test _apple

Output-

```
Console   Terminal ×   Background Jobs ×
R 4.5.2 · ~/
> apple <- read.csv(file.choose(), sep = ",", header = TRUE, stringsAsFactors = FALSE)
> colnames(apple) <- trimws(colnames(apple))
> summary(apple)
       C1
 Min.   : 48.90
 1st Qu.: 87.50
 Median : 96.40
 Mean   : 97.52
 3rd Qu.:108.50
 Max.   :135.00
> apple$C1 <- as.numeric(apple$C1)
> t.test(apple$C1, alternative = "less", mu = 97)

        One Sample t-test

data:  apple$C1
t = 0.14561, df = 27, p-value = 0.5573
alternative hypothesis: true mean is less than 97
95 percent confidence interval:
    -Inf 103.621
sample estimates:
mean of x
 97.52143
```

2)Sample t test: Male and Female salary

```
Salary <- read.csv(file.choose(), sep = ",", header = TRUE)

summary(Salary)

t.test(Salary$MALES, Salary$FEMALES,
       alternative = "two.sided",
       var.equal = TRUE)
```

2).CSV file to use-

2 sample t test_malefemalesalary - 2 sample t test_malefemalesalary

Output-

```
1:1   (Top Level) ÷

Console   Terminal ×   Background Jobs ×

R 4.5.2 · ~/

> Salary <- read.csv(file.choose(), sep = ",", header = TRUE)
>
> summary(Salary)
     MALES           FEMALES
 Min.   :25.30   Min.   :25.50
 1st Qu.:30.90   1st Qu.:31.20
 Median :33.75   Median :33.25
 Mean   :33.43   Mean   :33.20
 3rd Qu.:35.25   3rd Qu.:34.95
 Max.   :40.50   Max.   :41.20
>
> t.test(Salary$MALES, Salary$FEMALES,
+        alternative = "two.sided",
+        var.equal = TRUE)

        Two Sample t-test

data:  Salary$MALES and Salary$FEMALES
t = 0.21514, df = 42, p-value = 0.8307
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 -1.942752  2.406388
sample estimates:
mean of x mean of y
 33.43182  33.20000
```

3)**Paired t test: Poverty level**

```
poverty <- read.csv(file.choose(), sep = ",", header = TRUE)

t.test(poverty$X1, poverty$X2,
       alternative = "greater",
       paired = TRUE)
```

3).CSV file to use-

paired_t_test_povertylevel - paired_t_test_povertylevel

Output-

```
1:1    (Top Level) ÷
Console   Terminal ×   Background Jobs ×
R 4.5.2 · ~/
> poverty <- read.csv(file.choose(), sep = ",", header = TRUE)
>
> t.test(poverty$X1, poverty$X2,
+        alternative = "greater",
+        paired = TRUE)

        Paired t-test

data:  poverty$X1 and poverty$X2
t = -2.339, df = 50, p-value = 0.9883
alternative hypothesis: true mean difference is greater than 0
95 percent confidence interval:
 -1.511189       Inf
sample estimates:
mean difference
     -0.8803922

>
```

4)Paired t test: MIS Report

```
time1 <- read.csv(file.choose(), sep = ",", header = TRUE)

timel <- time1

t.test(timel$time_before, timel$time_after,
       alternative = "less",
       paired = TRUE)
```

4).CSV file to use-

PAIRED t TEST - PAIRED t TEST

Output

```
1:1    (Top Level) ⇕
Console   Terminal ×   Background Jobs ×
R ▾ R 4.5.2 · ~/
> time1 <- read.csv(file.choose(), sep = ",", header = TRUE)
>
> timel <- time1
>
> t.test(timel$time_before, timel$time_after,
+        alternative = "less",
+        paired = TRUE)

        Paired t-test

data:  timel$time_before and timel$time_after
t = 8.2295, df = 14, p-value = 1
alternative hypothesis: true mean difference is less than 0
95 percent confidence interval:
     -Inf 10.11687
sample estimates:
mean difference
       8.333333

>
```

5) **Correlation test**

```
cor <- read.csv(file.choose(), sep = ",", header = TRUE)

summary(cor)

cor.test(cor$aptitude, cor$job_prof,
        alternative = "two.sided",
        method = "pearson")
```

5).CSV file to use-

correlation test (1)

Output

```
Console   Terminal ×   Background Jobs ×
R 4.5.2 · ~/
> cor <- read.csv(file.choose(), sep = ",", header = TRUE)
>
> summary(cor)
   Empcode             aptitude        job_prof
 Length:25          Min.   : 62.0   Min.   : 58.0
 Class :character   1st Qu.: 91.0   1st Qu.: 78.0
 Mode  :character   Median :104.0   Median : 94.0
                    Mean   :103.4   Mean   : 92.2
                    3rd Qu.:112.0   3rd Qu.:109.0
                    Max.   :150.0   Max.   :127.0
>
> cor.test(cor$aptitude, cor$job_prof,
+          alternative = "two.sided",
+          method = "pearson")

        Pearson's product-moment correlation

data:  cor$aptitude and cor$job_prof
t = 2.8769, df = 23, p-value = 0.008517
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 0.1497097 0.7558981
sample estimates:
      cor
0.5144107
```

6)**Variances test**

```
var <- read.csv(file.choose(), sep = ",", header = TRUE)

summary(var)

var.test(var$time_g1, var$time_g2,
         alternative = "two.sided")
```

6).CSV file to use-

variances (1)

Output

```
Console   Terminal ×   Background Jobs ×
R 4.5.2 · ~/
> var <- read.csv(file.choose(), sep = ",", header = TRUE)
>
> summary(var)
    time_g1          time_g2
 Min.   : 85.00   Min.   : 83.00
 1st Qu.: 89.50   1st Qu.: 90.50
 Median : 94.50   Median : 94.00
 Mean   : 93.58   Mean   : 93.07
 3rd Qu.: 95.50   3rd Qu.: 95.00
 Max.   :105.00   Max.   :102.00
 NA's   :2
>
> var.test(var$time_g1, var$time_g2,
+          alternative = "two.sided")

        F test to compare two variances

data:  var$time_g1 and var$time_g2
F = 1.5434, num df = 11, denom df = 13, p-value = 0.4524
alternative hypothesis: true ratio of variances is not equal to 1
95 percent confidence interval:
 0.4826988 5.2348866
sample estimates:
ratio of variances
          1.543428
```

Prac 3-
Codes-

1) ONE WAY ANOVA
CODE-

```
data1 <- read.csv(file.choose(), sep = ",", header = TRUE)

names(data1)
summary(data1)
head(data1)

anv <- aov(satindex ~ dept, data = data1)

summary(anv)
```

1).CSV file to use-

One way anova

Output

```
> data1 <- read.csv(file.choose(), sep = ",", header = TRUE)
>
> names(data1)
[1] "satindex" "dept"
> summary(data1)
    satindex         dept
 Min.   :51.00   Length:37
 1st Qu.:59.00   Class :character
 Median :66.00   Mode  :character
 Mean   :65.59
 3rd Qu.:71.00
 Max.   :77.00
> head(data1)
  satindex    dept
1       75 FINANCE
2       56 FINANCE
3       72 FINANCE
4       59 FINANCE
5       62 FINANCE
6       66 FINANCE
>
> anv <- aov(satindex ~ dept, data = data1)
>
> summary(anv)
            Df Sum Sq Mean Sq F value Pr(>F)
dept         2  220.1  110.03   2.308  0.115
Residuals   34 1620.9   47.67
```

2) TWO WAY ANOVA

CODE-

```
data2 <- read.csv(file.choose(), sep = ",", header = TRUE)

names(data2)
summary(data2)

anv1 <- aov(satindex ~ dept + exp + dept:exp, data = data2)

summary(anv1)
```

2).CSV file to use-

Two way anova

Output-

```
Console   Terminal ×   Background Jobs ×
R 4.5.2 · ~/
> data2 <- read.csv(file.choose(), sep = ",", header = TRUE)
>
> names(data2)
[1] "satindex" "dept"     "exp"
> summary(data2)
    satindex         dept               exp
 Min.   :51.00   Length:36          Length:36
 1st Qu.:59.00   Class :character   Class :character
 Median :66.00   Mode  :character   Mode  :character
 Mean   :65.31
 3rd Qu.:71.00
 Max.   :77.00
>
> anv1 <- aov(satindex ~ dept + exp + dept:exp, data = data2)
>
> summary(anv1)
            Df Sum Sq Mean Sq F value Pr(>F)
dept         2  164.2   82.11   1.679  0.204
exp          1   78.0   78.03   1.595  0.216
dept:exp     2   20.2   10.11   0.207  0.814
Residuals   30 1467.2   48.91
>
```

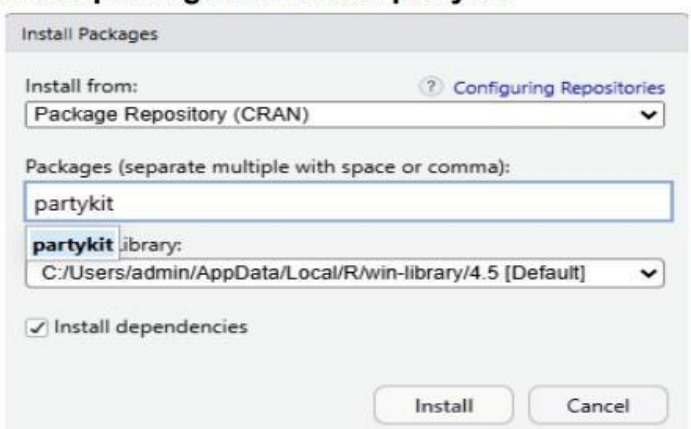
Prac 4 -
1)Decision Tree
Code-

```
library(MASS)
library(rpart)
library(partykit)

data(biopsy)

biopsy <- biopsy[, -1]

names(biopsy) <- c("thick", "u.size", "u.shape", "adhsn",
                   "s.size", "nucl", "chrom", "n.nuc",
                   "mit", "class")

biopsy.v2 <- na.omit(biopsy)

set.seed(123)
ind <- sample(2, nrow(biopsy.v2),
              replace = TRUE,
              prob = c(0.7, 0.3))

biop.train <- biopsy.v2[ind == 1, ]
biop.test  <- biopsy.v2[ind == 2, ]

str(biop.test$class)

set.seed(123)
tree.biop <- rpart(class ~ .,
                   data = biop.train,
                   method = "class")

print(tree.biop$cptable)

cp <- tree.biop$cptable[which.min(tree.biop$cptable[, "xerror"]), "CP"]

prune.tree.biop <- prune(tree.biop, cp = cp)

plot(as.party(tree.biop))
```

```
plot(as.party(prune.tree.biop))

rparty.test <- predict(prune.tree.biop,
                       newdata = biop.test,
                       type = "class")

table(rparty.test, biop.test$class)
```

After the code install a package-

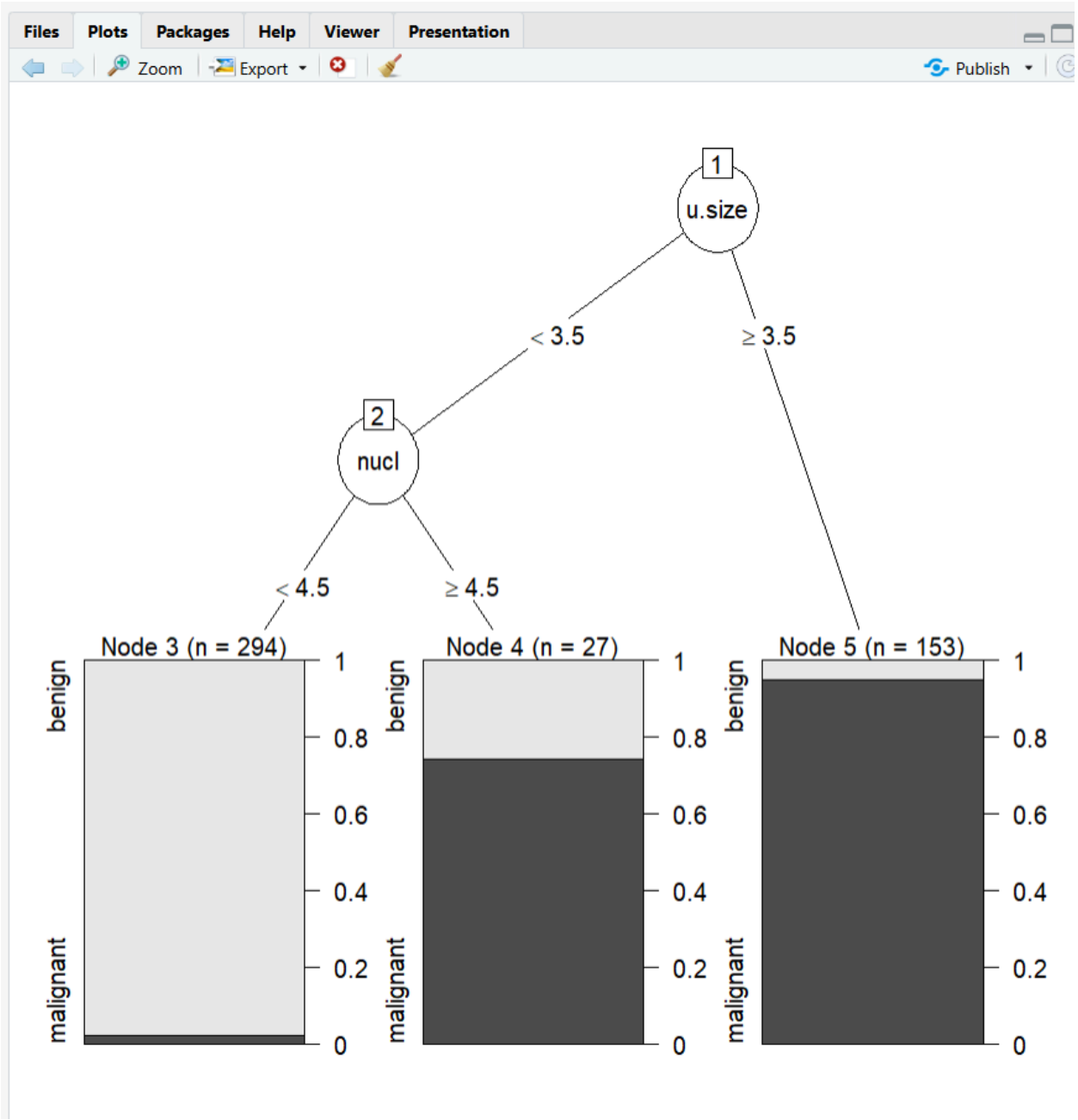
Go to package and install partykit



Output-

```
        C:\Users\Aditya\AppData\Local\Temp\Rtmp4EGNJf\downloaded_packages
> library(MASS)
> library(rpart)
> library(partykit)

Loading required package: grid
Loading required package: libcoin
Loading required package: mvtnorm

>
> data(biopsy)
>
> biopsy <- biopsy[, -1]
>
> names(biopsy) <- c("thick", "u.size", "u.shape", "adhsn",
+                    "s.size", "nucl", "chrom", "n.nuc",
+                    "mit", "class")
>
> biopsy.v2 <- na.omit(biopsy)
>
> set.seed(123)
> ind <- sample(2, nrow(biopsy.v2),
+               replace = TRUE,
+               prob = c(0.7, 0.3))
>
> biop.train <- biopsy.v2[ind == 1, ]
> biop.test  <- biopsy.v2[ind == 2, ]
>
> str(biop.test$class)
 Factor w/ 2 levels "benign","malignant": 1 1 1 1 1 2 1 2 1 1 ...
>
> set.seed(123)
> tree.biop <- rpart(class ~ .,
+                    data = biop.train,
+                    method = "class")
>
> print(tree.biop$cptable)
          CP nsplit rel error    xerror       xstd
1 0.79651163      0 1.0000000 1.0000000 0.06086254
2 0.07558140      1 0.2034884 0.2616279 0.03710371
3 0.0116279l      2 0.1279070 0.1511628 0.02882093
4 0.01000000      3 0.1162791 0.1511628 0.02882093
>
> cp <- tree.biop$cptable[which.min(tree.biop$cptable[, "xerror"]), "CP"]
>
> prune.tree.biop <- prune(tree.biop, cp = cp)
>
> plot(as.party(tree.biop))
> plot(as.party(prune.tree.biop))
>
> rparty.test <- predict(prune.tree.biop,
+                        newdata = biop.test,
+                        type = "class")
>
> table(rparty.test, biop.test$class)

rparty.test benign malignant
  benign       136         3
  malignant      6        64
>
```

Plot output-


Files
Plots
Packages
Help
Viewer
Presentation
Zoom
Export
Publish
1
u.size
< 3.5
≥ 3.5
2
nucl
< 4.5
≥ 4.5
Node 3 (n = 294)
Node 4 (n = 27)
Node 5 (n = 153)
benign
malignant
1
0.8
0.6
0.4
0.2
0

Prac 6-

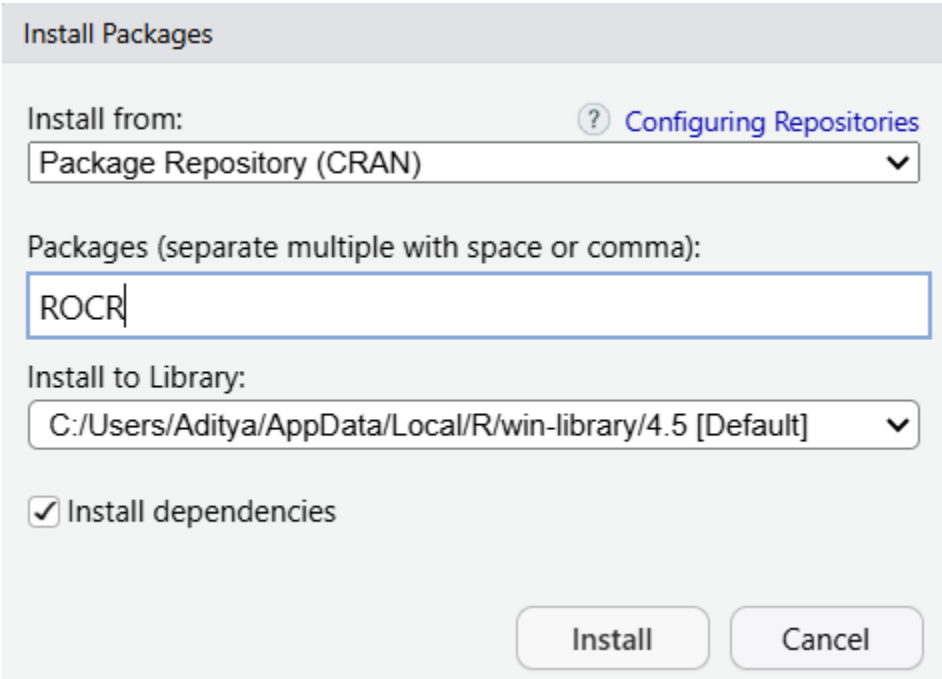
Logistic Regression-

Code-

```
library(MASS)
data(biopsy)

head(biopsy)
str(biopsy)

biopsy$ID = NULL

names(biopsy) = c("thick","usize","ushape","adhsn",
                  "celsiz","nucl","chrom",
                  "nuclus","mit","class")

colSums(is.na(biopsy))

biopsy1 = na.omit(biopsy)

set.seed(123)
ind = sample(2, nrow(biopsy1), replace = TRUE, prob = c(0.7, 0.3))

train = biopsy1[ind == 1, ]
test  = biopsy1[ind == 2, ]

fullfit = glm(class ~ ., family = binomial, data = train)
summary(fullfit)

test$prob = predict(fullfit, newdata = test, type = "response")

test$predict = rep("benign", nrow(test))
test$predict[test$prob < 0.5] = "malignant"

table(test$predict, test$class)

mean(test$predict == test$class)

library(ROCR)
```

```
ROCRpred = prediction(test$prob, test$class)
ROCRperf = performance(ROCRpred, "tpr", "fpr")

plot(ROCRperf, colorize = TRUE)
```

After the code install package



Output-

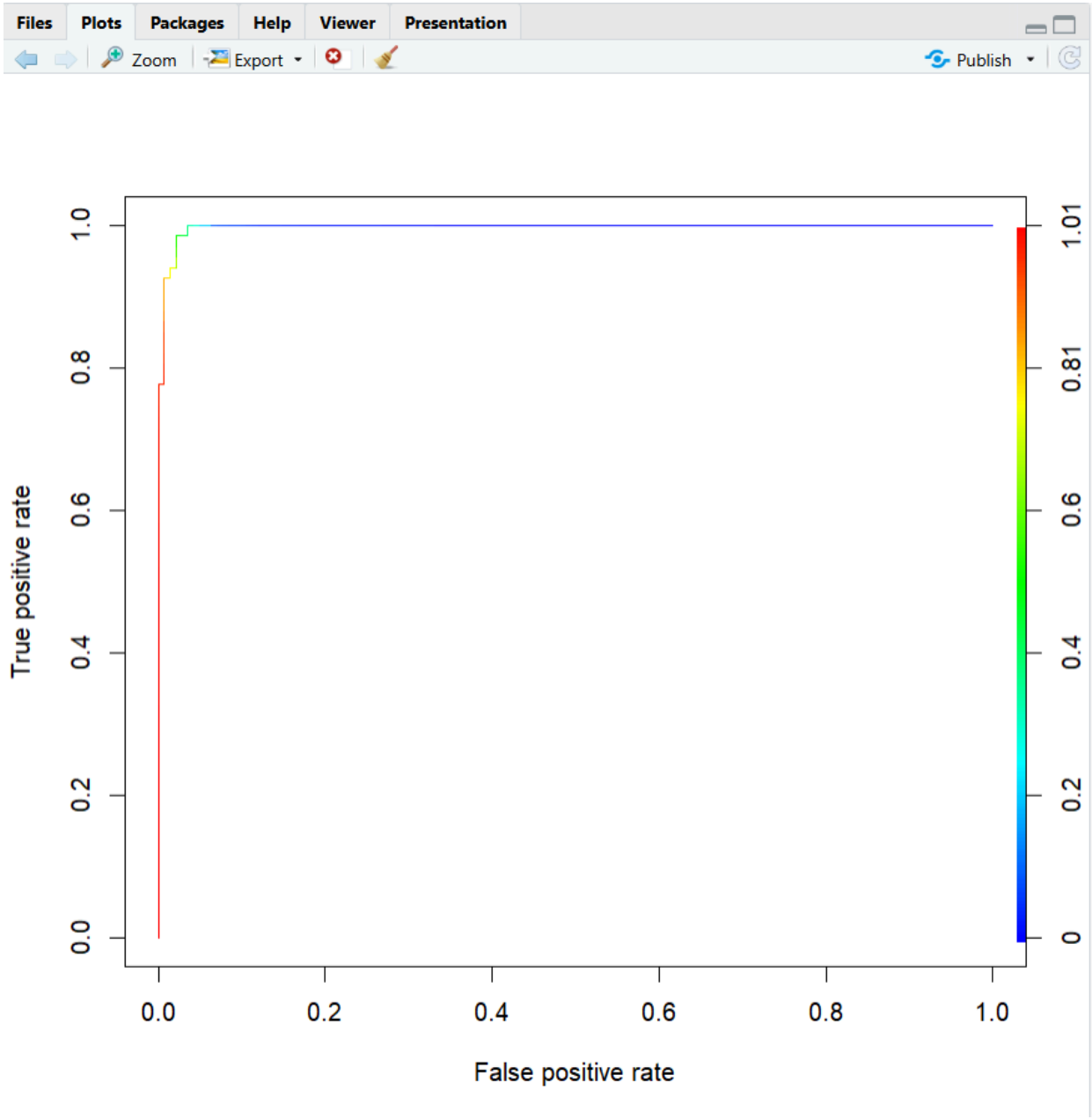
Console Terminal Background Jobs

R 4.5.2 · ~/

```
> library(MASS)
> data(biopsy)
>
> head(biopsy)
       ID V1 V2 V3 V4 V5 V6 V7 V8 V9     class
1 1000025  5  1  1  1  2  1  3  1  1    benign
2 1002945  5  4  4  5  7 10  3  2  1    benign
3 1015425  3  1  1  1  2  2  3  1  1    benign
4 1016277  6  8  8  1  3  4  3  7  1    benign
5 1017023  4  1  1  3  2  1  3  1  1    benign
6 1017122  8 10 10  8  7 10  9  7  1 malignant
> str(biopsy)
'data.frame':   699 obs. of  11 variables:
 $ ID   : chr  "1000025" "1002945" "1015425" "1016277" ...
 $ V1   : int  5 5 3 6 4 8 1 2 2 4 ...
 $ V2   : int  1 4 1 8 1 10 1 1 1 2 ...
 $ V3   : int  1 4 1 8 1 10 1 2 1 1 ...
 $ V4   : int  1 5 1 1 3 8 1 1 1 1 ...
 $ V5   : int  2 7 2 3 2 7 2 2 2 2 ...
 $ V6   : int  1 10 2 4 1 10 10 1 1 1 ...
 $ V7   : int  3 3 3 3 3 9 3 3 1 2 ...
 $ V8   : int  1 2 1 7 1 7 1 1 1 1 ...
 $ V9   : int  1 1 1 1 1 1 1 1 5 1 ...
 $ class: Factor w/ 2 levels "benign","malignant": 1 1 1 1 1 2 1 1 1 1 ...
>
> biopsy$ID = NULL
>
> names(biopsy) = c("thick","usize","ushape","adhsn",
+                   "celsiz","nucl","chrom",
+                   "nuclus","mit","class")
>
> colSums(is.na(biopsy))
 thick  usize ushape  adhsn celsiz   nucl  chrom nuclus    mit  class
     0      0      0      0      0     16      0      0      0      0
>
> biopsy1 = na.omit(biopsy)
>
> set.seed(123)
> ind = sample(2, nrow(biopsy1), replace = TRUE, prob = c(0.7, 0.3))
>
> train = biopsy1[ind == 1, ]
> test  = biopsy1[ind == 2, ]
>
> fullfit = glm(class ~ ., family = binomial, data = train)
> summary(fullfit)

Call:
glm(formula = class ~ ., family = binomial, data = train)

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept)  -9.4293     1.2273  -7.683 1.55e-14 ***
thick         0.5252     0.1601   3.280 0.001039 **
usize        -0.1045     0.2446  -0.427 0.669165
ushape        0.2798     0.2526   1.108 0.268044
adhsn         0.3086     0.1738   1.776 0.075722 .
celsiz        0.2866     0.2074   1.382 0.167021
nucl          0.4057     0.1213   3.344 0.000826 ***
chrom         0.2737     0.2174   1.259 0.208006
nuclus        0.2244     0.1373   1.635 0.102126
mit           0.4296     0.3393   1.266 0.205402
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

(Dispersion parameter for binomial family taken to be 1)
```

```
(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 620.989  on 473  degrees of freedom
Residual deviance:  78.373  on 464  degrees of freedom
AIC: 98.373

Number of Fisher Scoring iterations: 8

>
> test$prob = predict(fullfit, newdata = test, type = "response")
>
> test$predict = rep("benign", nrow(test))
> test$predict[test$prob < 0.5] = "malignant"
>
> table(test$predict, test$class)

           benign malignant
  benign        3        65
  malignant   139         2
>
> mean(test$predict == test$class)
[1] 0.02392344
>
> library(ROCR)
>
> ROCRpred = prediction(test$prob, test$class)
> ROCRperf = performance(ROCRpred, "tpr", "fpr")
>
> plot(ROCRperf, colorize = TRUE)
>
```

Plot Output-


Files
Plots
Packages
Help
Viewer
Presentation
Zoom
Export
Publish
True positive rate
False positive rate
0.0
0.2
0.4
0.6
0.8
1.0
0
0.2
0.4
0.6
0.81
1.01

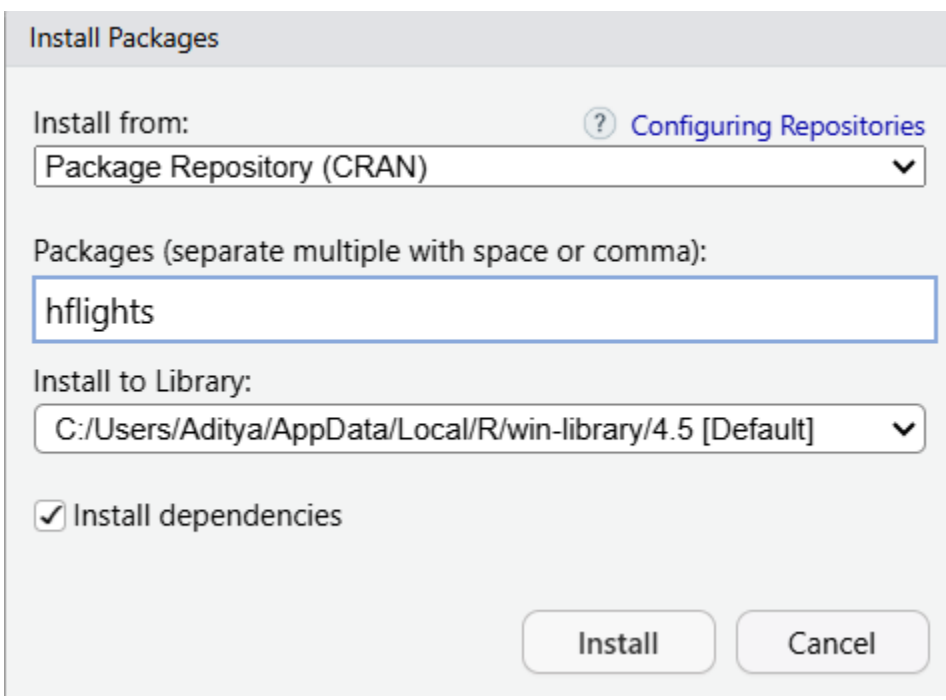
Prac 7

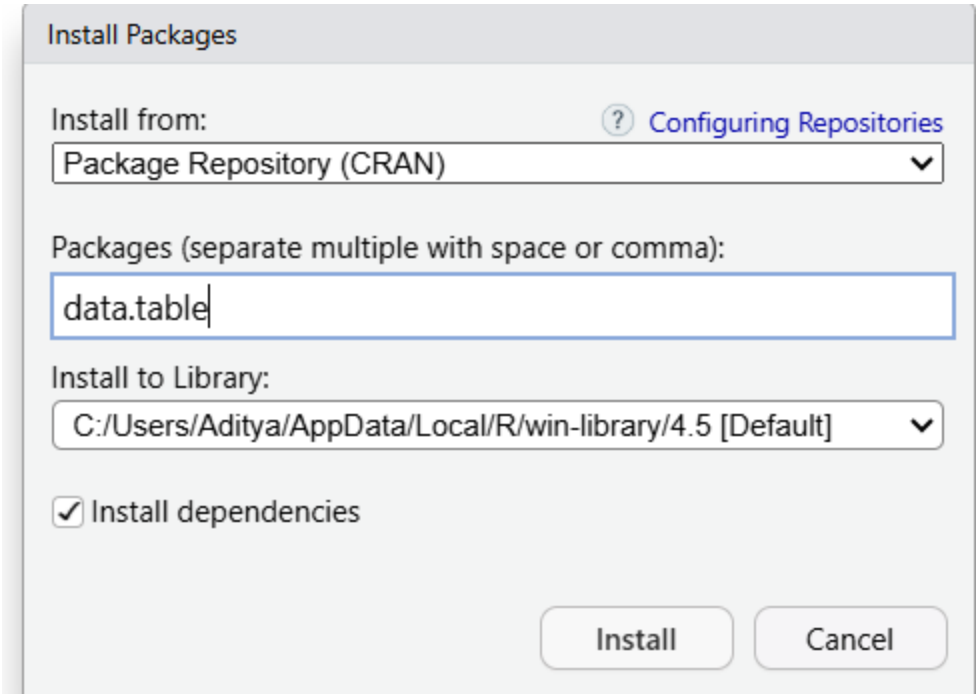
Time Series Forecasting

Code-

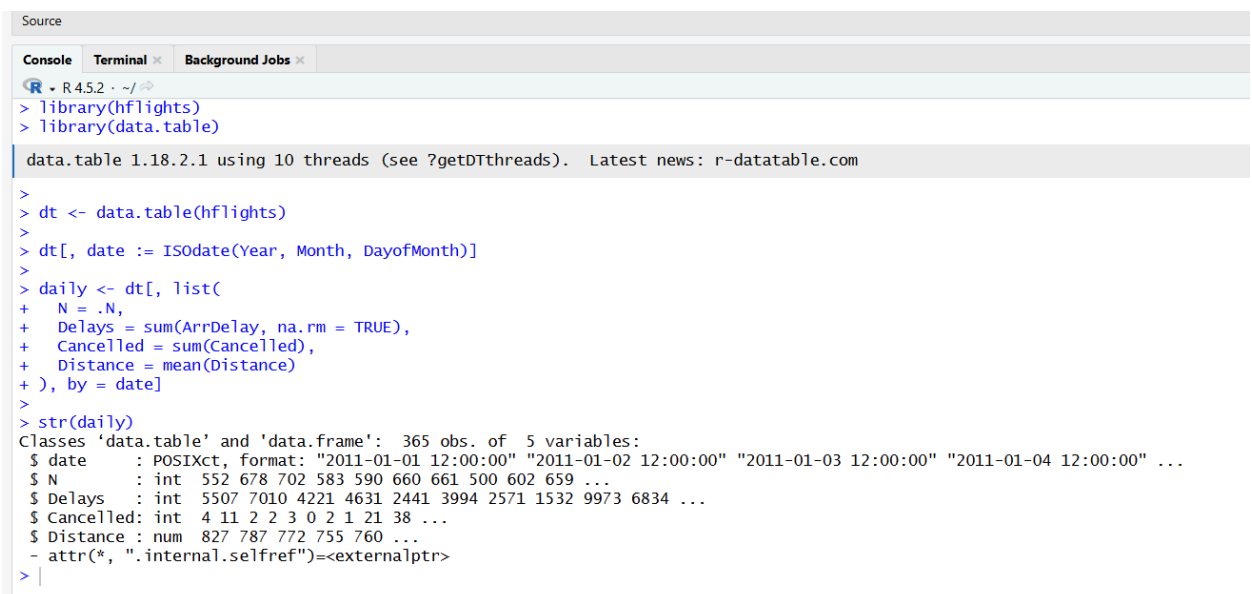
```
library(hflights)
library(data.table)

dt <- data.table(hflights)

dt[, date := ISOdate(Year, Month, DayofMonth)]

daily <- dt[, list(
  N = .N,
  Delays = sum(ArrDelay, na.rm = TRUE),
  Cancelled = sum(Cancelled),
  Distance = mean(Distance)
), by = date]

str(daily)
```

After the code install packages -





OUTPUT-

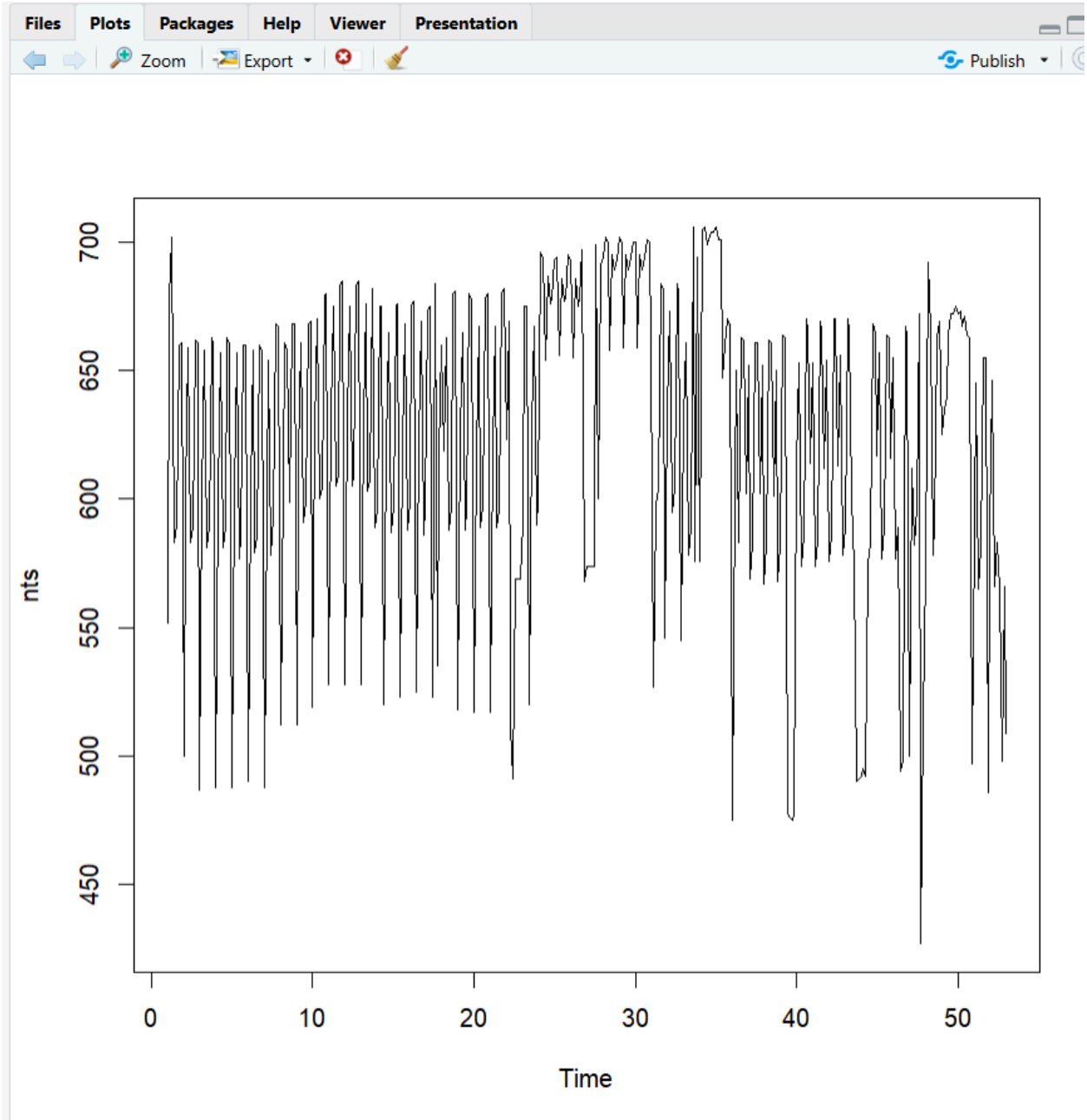


After this Output type in the Output This line -

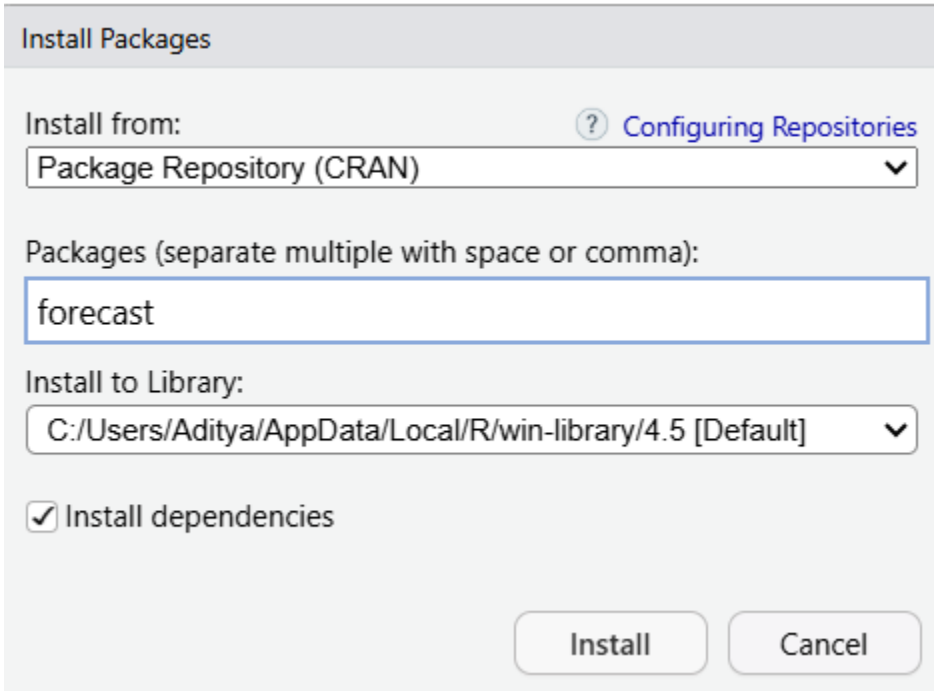
nts <- ts(daily$N, frequency = 7)

plot(nts)

It will show you plot output-



After this Plot output install a package

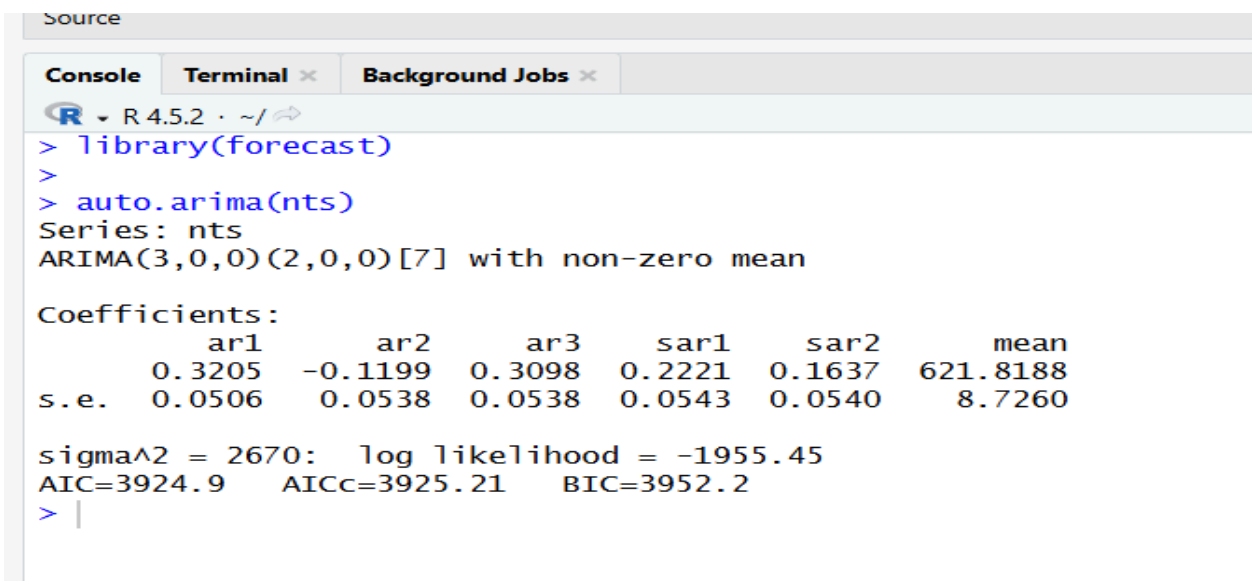


After installing the package again type in the output this line-

library(forecast)

auto.arima(nts)

This will show you different output-



After this again type in the output this line-

auto.arima(nts, approximation = FALSE)

This will give you output-

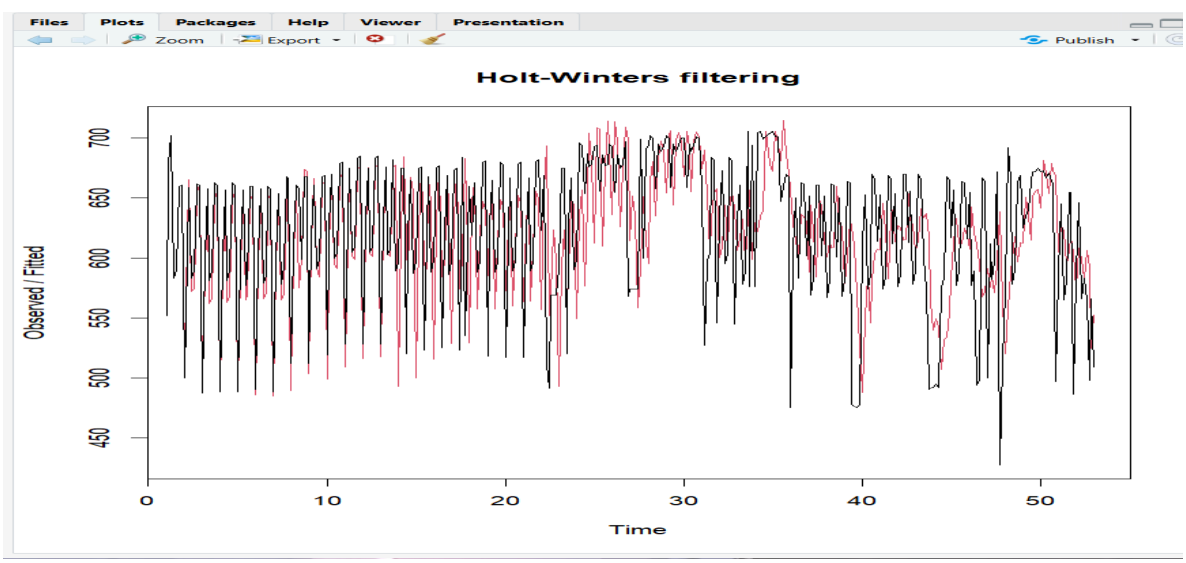
```
Console   Terminal ×   Background Jobs ×
R 4.5.2 · ~/
> auto.arima(nts, approximation = FALSE)
Series: nts
ARIMA(0,0,4)(2,0,0)[7] with non-zero mean

Coefficients:
         ma1      ma2     ma3     ma4    sar1    sar2      mean
      0.3257  -0.0311  0.2211  0.2364  0.2801  0.1392  621.9295
s.e.  0.0531   0.0531  0.0496  0.0617  0.0534  0.0557    7.9372

sigma^2 = 2683:  log likelihood = -1955.83
AIC=3927.66   AICc=3928.07   BIC=3958.86
>
```

After this again type in the output this line-

fit <- HoltWinters(nts)

plot(fit)

This will give you the different plot output-



Again type in the output this line-

library(forecast)

forecast(fit)

This will show you the output-

```
Console   Terminal ×   Background Jobs ×
R 4.5.2 · ~/
> library(forecast)
>
> forecast(fit)
         Point Forecast    Lo 80    Hi 80    Lo 95    Hi 95
53.14286       574.5950 507.4414 641.7485 471.8925 677.2974
53.28571       546.6390 477.2104 616.0677 440.4570 652.8210
53.42857       542.6120 470.9407 614.2833 433.0003 652.2238
53.57143       560.5536 486.6690 634.4382 447.5568 673.5503
53.71429       550.2924 474.2208 626.3640 433.9509 666.6339
53.85714       524.2956 446.0608 602.5304 404.6458 643.9454
54.00000       530.3396 449.9632 610.7160 407.4145 653.2647
54.14286       569.4380 484.6054 654.2707 439.6977 699.1783
54.28571       541.4821 454.6017 628.3624 408.6101 674.3541
54.42857       537.4551 448.5402 626.3700 401.4714 673.4387
54.57143       555.3966 464.4592 646.3341 416.3198 694.4735
54.71429       545.1355 452.1864 638.0845 402.9821 687.2888
54.85714       519.1386 424.1879 614.0894 373.9240 664.3532
55.00000       525.1826 428.2393 622.1259 376.9207 673.4446
>
```

Again type in the output this line-

f <- forecast(HoltWinters(nts), 5)

f

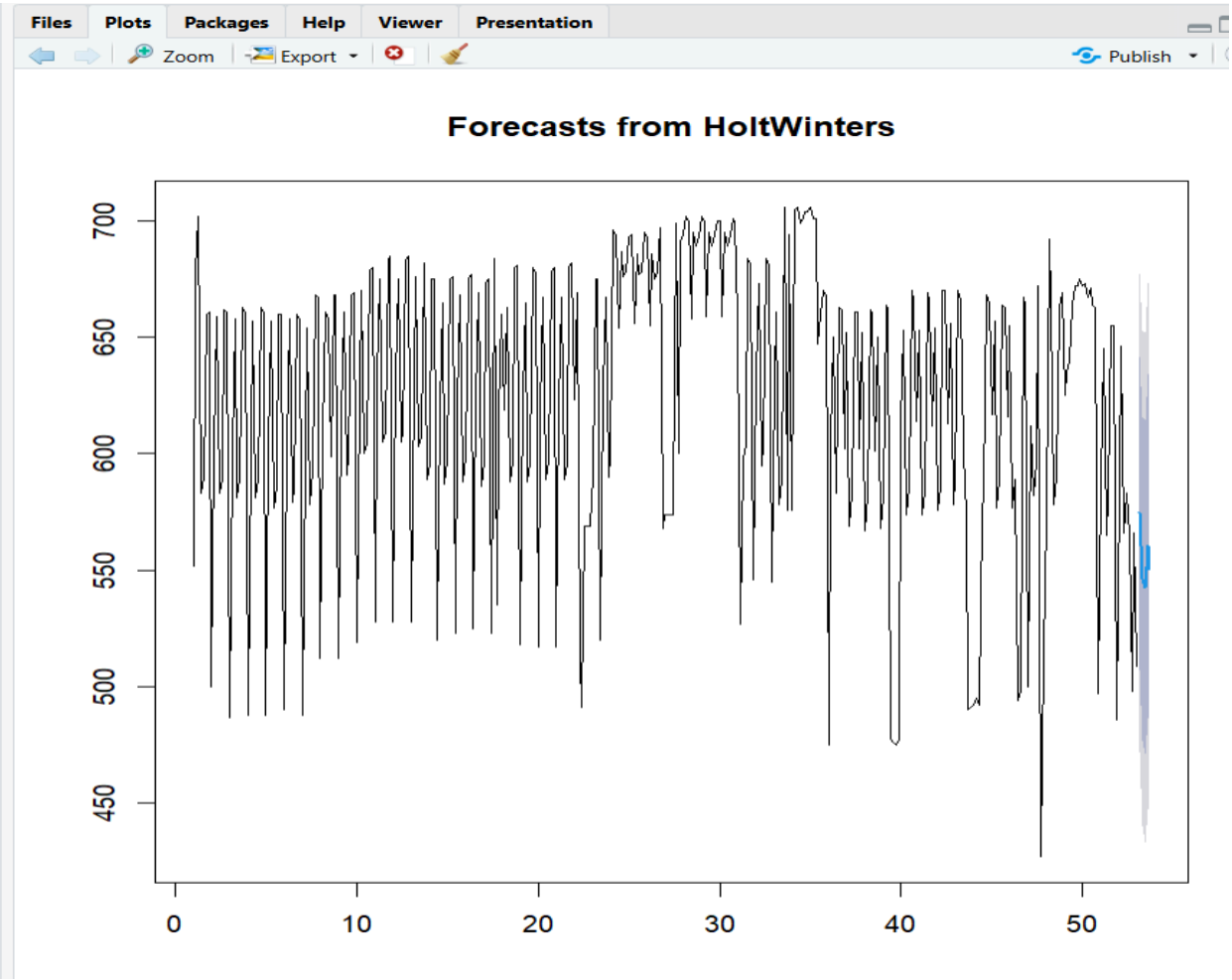
This will show you the output-

```
Console   Terminal ×   Background Jobs ×
R 4.5.2 · ~/
> f <- forecast(HoltWinters(nts), 5)
> f
         Point Forecast    Lo 80    Hi 80    Lo 95    Hi 95
53.14286       574.5950 507.4414 641.7485 471.8925 677.2974
53.28571       546.6390 477.2104 616.0677 440.4570 652.8210
53.42857       542.6120 470.9407 614.2833 433.0003 652.2238
53.57143       560.5536 486.6690 634.4382 447.5568 673.5503
53.71429       550.2924 474.2208 626.3640 433.9509 666.6339
>
```

After this again type in the output-

```
library(forecast)

nts <- ts(daily$N, frequency = 7)

fit <- HoltWinters(nts)

f <- forecast(fit, 5)

plot(f)
```

This will give you the plot output-



Again type in the output this line-

cts <- ts(daily$Cancelled)

fit <- auto.arima(cts)

auto.arima(cts)

This will give you the output-

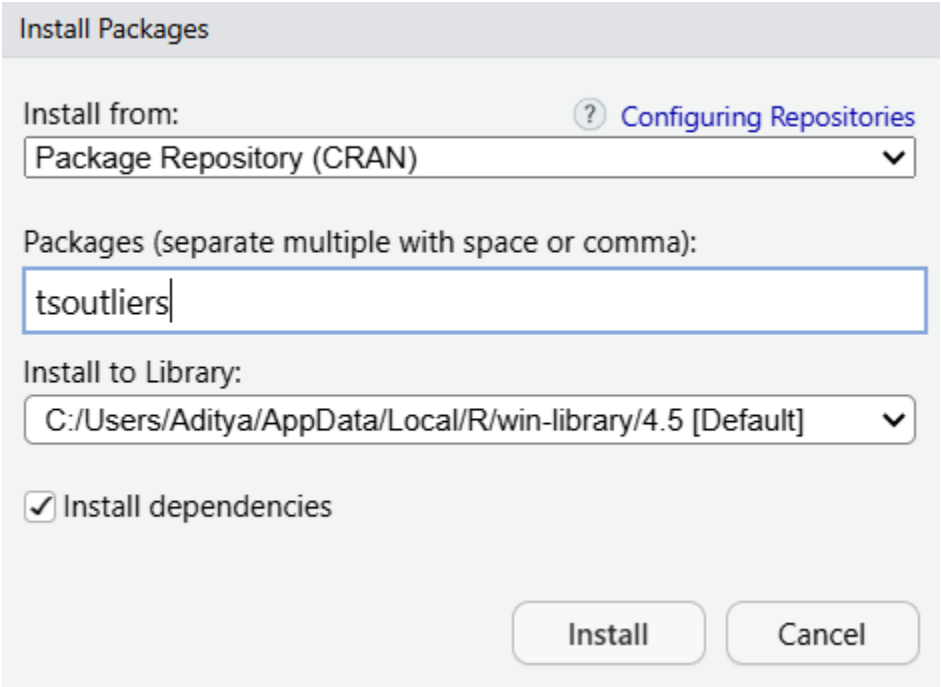
Source

Console | Terminal | Background Jobs

R 4.5.2 · ~/

```
> cts <- ts(daily$Cancelled)
>
> fit <- auto.arima(cts)
>
> auto.arima(cts)
Series: cts
ARIMA(1,1,2)

Coefficients:
          ar1      ma1      ma2
      -0.2601  -0.1787  -0.7752
s.e.   0.0969   0.0746   0.0640

sigma^2 = 544.3:  log likelihood = -1662.95
AIC=3333.9   AICc=3334.01   BIC=3349.49
> |
```

After this install a package-



Then again type in the output this line -

```
library(tsoutliers)

outliers <- tso(
  cts,
  tsmethod = "arima",
  args.tsmethod = list(order = c(1, 1, 2))
)

outliers
```

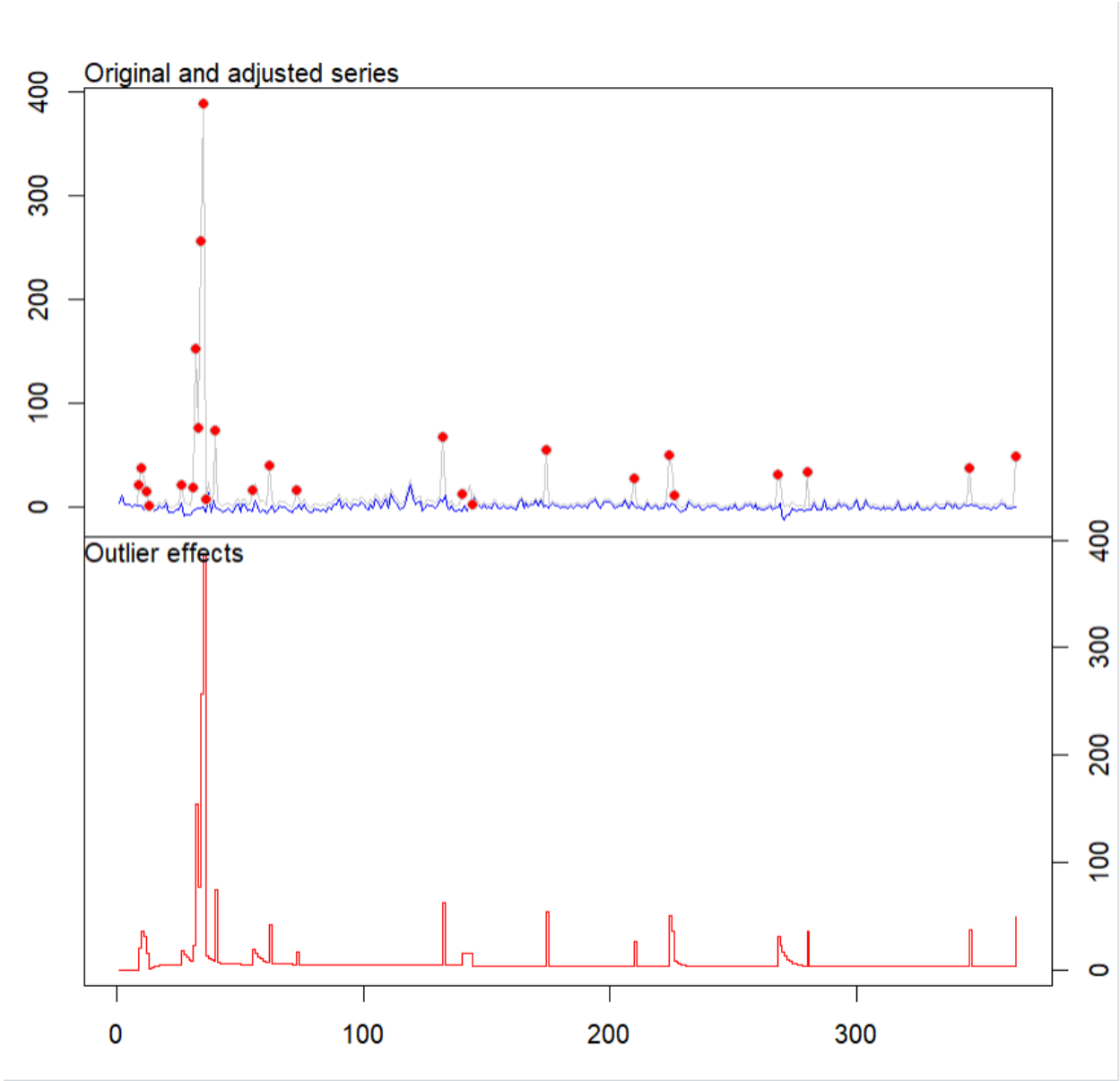
OUTPUT-

```
Call:
list(method = NULL)

Coefficients:
         ar1      ma1      ma2      LS9     LS13     AO31     TC32     AO33     TC34     AO35     TC36     AO40     AO62    AO132    AO143    AO174
      0.0810  -0.7342  -0.1995  21.2345  -20.5371  15.4537  147.6625  -31.7795  179.3687  209.6428  -114.9636  65.2175  35.1779  55.8650  14.9937  51.9155
s.e.  0.1898   0.1873   0.1709   2.7151    2.5141   3.6687    3.8603    3.9494    4.1779    4.0665     3.7857   3.5800   3.4818   3.5495   3.5357   3.4915
       AO210    TC224    TC226    TC268    AO280    AO346    AO365
      22.7673  47.1131  -18.6309  26.6792  33.0172  34.4361  46.1150
s.e.   3.4729   3.4524    3.4248   3.3053   3.4775   3.4763   3.6968

sigma^2 estimated as 13.66:  log likelihood = -993.17,  aic = 2034.34

Outliers:
   type ind time coefhat   tstat
1    LS   9    9   21.23   7.821
2    LS  13   13  -20.54  -8.169
3    AO  31   31   15.45   4.212
4    TC  32   32  147.66  38.252
5    AO  33   33  -31.78  -8.047
6    TC  34   34  179.37  42.933
7    AO  35   35  209.64  51.553
8    TC  36   36 -114.96 -30.368
9    AO  40   40   65.22  18.217
10   AO  62   62   35.18  10.103
11   AO 132  132   55.87  15.739
12   AO 143  143   14.99   4.241
13   AO 174  174   51.92  14.869
14   AO 210  210   22.77   6.556
15   TC 224  224   47.11  13.646
16   TC 226  226  -18.63  -5.440
17   TC 268  268   26.68   8.072
18   AO 280  280   33.02   9.494
19   AO 346  346   34.44   9.906
20   AO 365  365   46.11  12.474
>
```

After this type this in output for plot output-

```
library(tsoutliers)

cts <- ts(daily$Cancelled)

outliers <- tso(
  cts,
  tsmethod = "arima",
  args.tsmethod = list(order = c(1, 1, 2))
)

plot(outliers)

plot(tso(ts(daily$Cancelled)))
```

OUTPUT-


Original and adjusted series
Outlier effects
0
100
200
300
400
0
100
200
300

PRAC 8

Principal component analysis

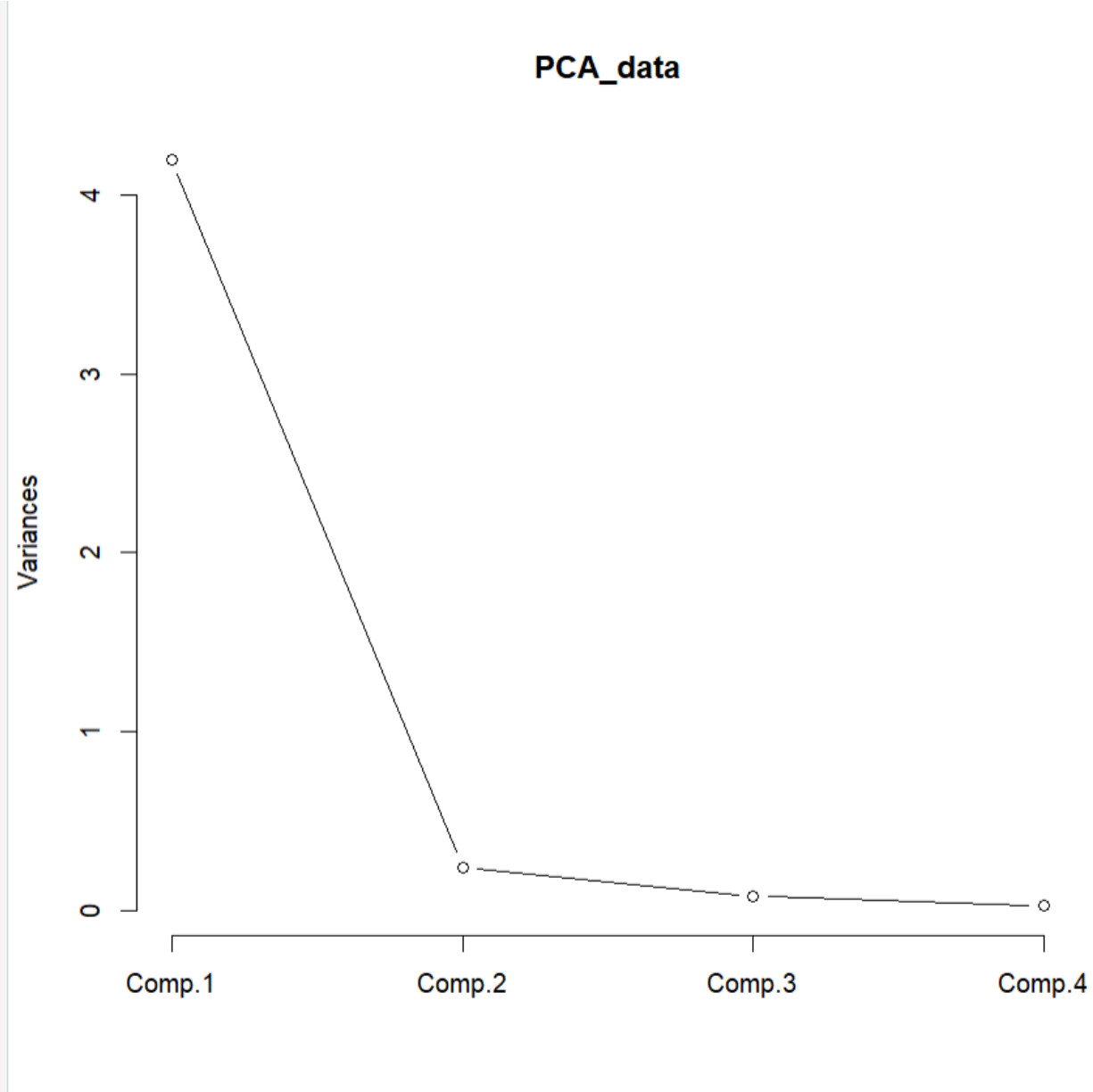
Code-

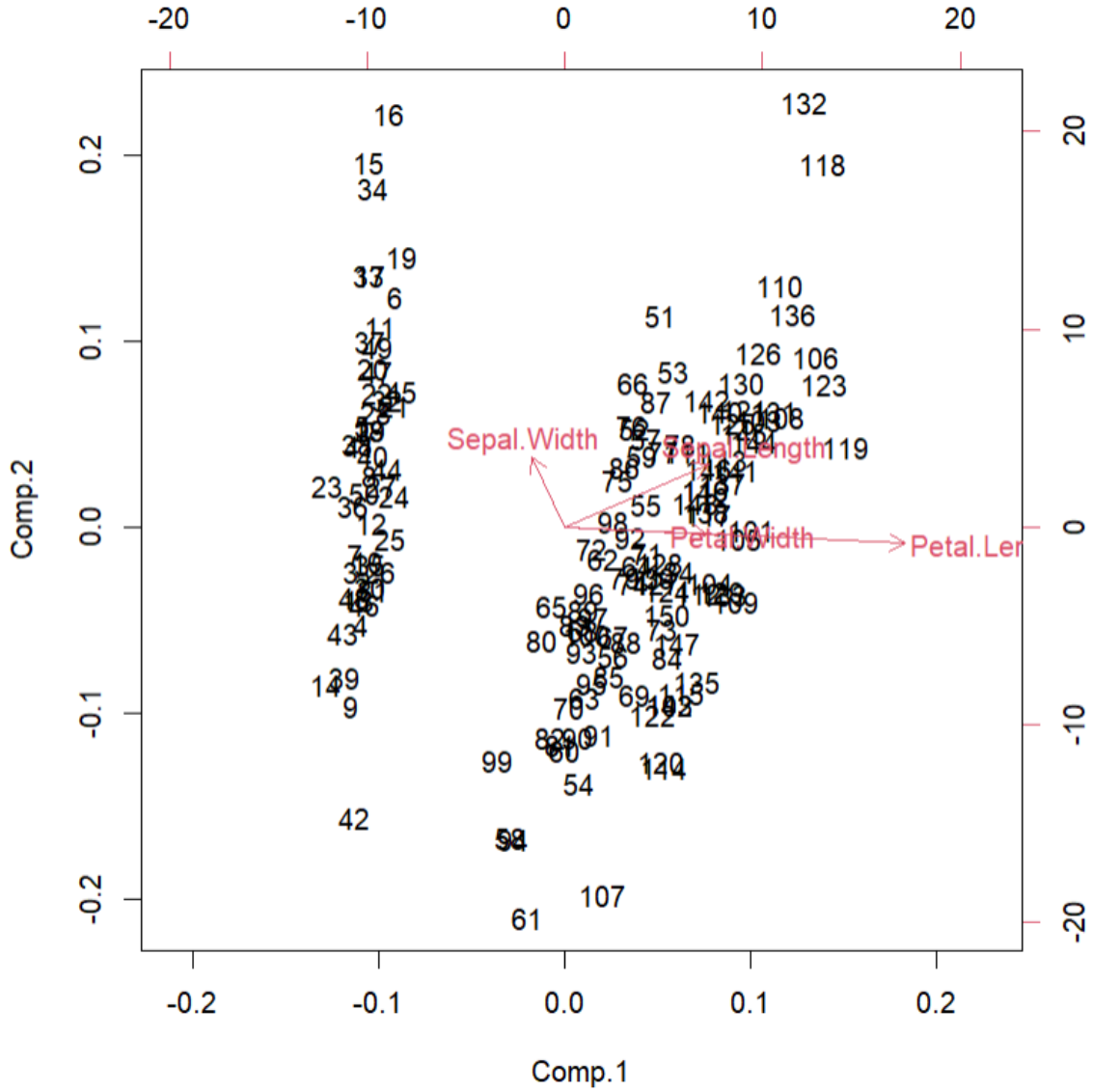
```
data("iris")

data_iris <- iris[, 1:4]

PCA_data <- princomp(data_iris, cor = FALSE)

PCA_data$loadings[, 1:4]

summary(PCA_data)

biplot(PCA_data)

screeplot(PCA_data, type = "lines")
data(iris)

data_iris <- iris[, 1:4]

PCA_data <- princomp(data_iris, cor = FALSE)

biplot(PCA_data)
```

OUTPUT-

```
Console   Terminal   Background Jobs
R 4.5.2 · ~/
> data("iris")
>
> data_iris <- iris[, 1:4]
>
> PCA_data <- princomp(data_iris, cor = FALSE)
>
> PCA_data$loadings[, 1:4]
              Comp.1       Comp.2      Comp.3     Comp.4
Sepal.Length  0.36138659  0.65658877  0.58202985  0.3154872
Sepal.Width  -0.08452251  0.73016143 -0.59791083 -0.3197231
Petal.Length  0.85667061 -0.17337266 -0.07623608 -0.4798390
Petal.Width   0.35828920 -0.07548102 -0.54583143  0.7536574
>
> summary(PCA_data)
Importance of components:
                          Comp.1     Comp.2     Comp.3      Comp.4
Standard deviation     2.0494032 0.49097143 0.27872586 0.153870700
Proportion of Variance 0.9246187 0.05306648 0.01710261 0.005212184
Cumulative Proportion  0.9246187 0.97768521 0.99478782 1.000000000
>
> biplot(PCA_data)
>
> screeplot(PCA_data, type = "lines")
> data(iris)
>
> data_iris <- iris[, 1:4]
>
> PCA_data <- princomp(data_iris, cor = FALSE)
>
> biplot(PCA_data)
>
```

Plot output-




Sepal.Width
Sepal.Length
Petal.Width
Petal.Le
Comp.1
Comp.2